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Web Application Crawling with Burp Suite

Mapping Application Structure Prior to Vulnerability Analysis — bWAPP on bee-box

Target Application	bWAPP (Buggy Web Application)
Target Environment	bee-box VM (VMware), 192.168.1.6
Tool Used	Burp Suite Community Edition v2026.4.3
Login Used	bee / bug (application test account)
Date	11 July 2026
Purpose	Independent practice — proxy-based web crawling & scoping

Note: This assessment was carried out in an isolated personal/virtual lab environment against intentionally vulnerable practice systems, as part of independent cybersecurity skill-building. No production systems, third-party assets, or real organizations were involved or affected.

1. Objective

The objective of this exercise was to perform web crawling on a deliberately vulnerable web application (bWAPP, hosted on the bee-box virtual machine) using Burp Suite, to map the application's structure, directories, and request/response traffic ahead of vulnerability analysis.

2. Lab Environment

Field	Detail
Tool Used	Burp Suite Community Edition v2026.4.3
Target Application	bWAPP (Buggy Web Application)
Target Environment	bee-box VM (VMware) — 192.168.1.6
Login Credentials	bee / bug (test account provided with the app)
Objective	Crawl the target web application and map its site structure
Date	11 July 2026

3. Procedure and Results

3.1 Setting Up the bee-box Target VM

The bee-box virtual machine (Ubuntu-based, with bWAPP pre-installed) was imported into VMware and powered on. A keyboard layout mismatch (AZERTY-style behaviour) was identified and corrected using setxkbmap us from the terminal.

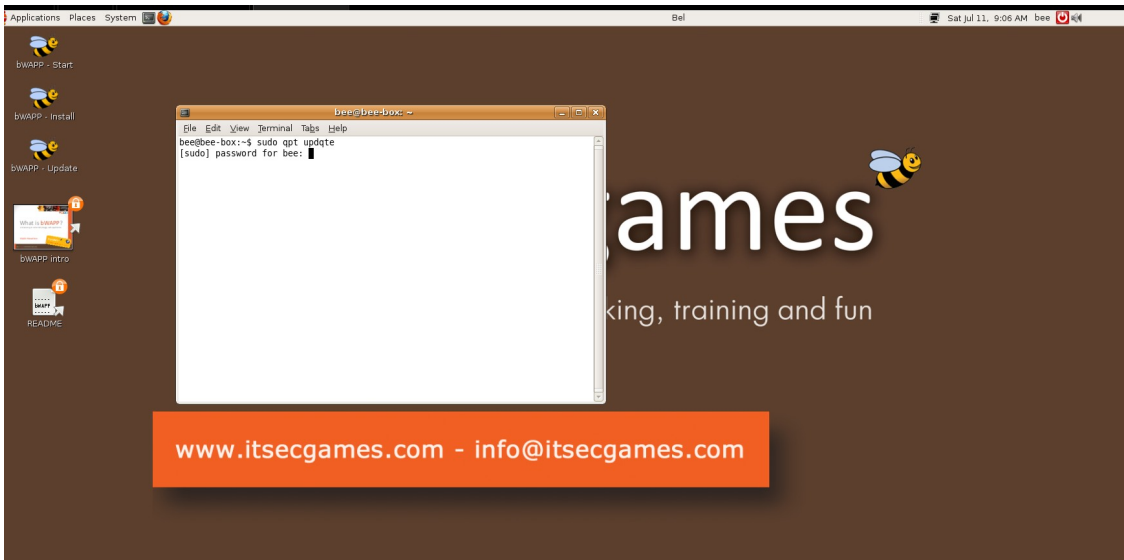


Figure 1: bee-box VM desktop with terminal open, showing bWAPP shortcuts and the sudo command prompt.

3.2 Installing Burp Suite

Burp Suite was downloaded on the attacking machine (Kali Linux). A Burp Suite Professional trial was attempted first but required a business email address, which was not available, so the exercise proceeded using Burp Suite Community Edition — fully capable of manual web crawling.

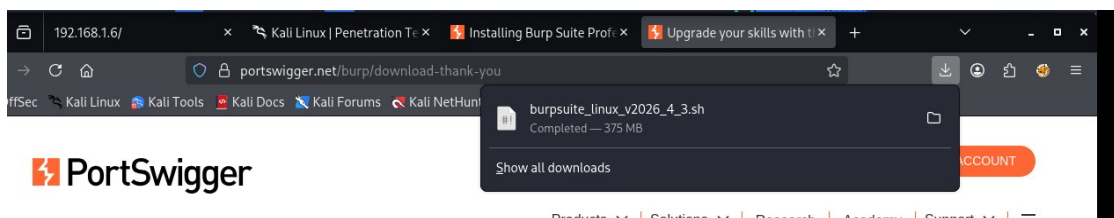


Figure 2: Burp Suite Professional installer (.sh) downloaded from PortSwigger.

3.3 Configuring the Proxy and Intercepting Traffic

Burp's proxy listener (127.0.0.1:8080) was configured and the browser routed through it. With Intercept switched on, initial requests to the target were captured, confirming the proxy chain was working correctly.

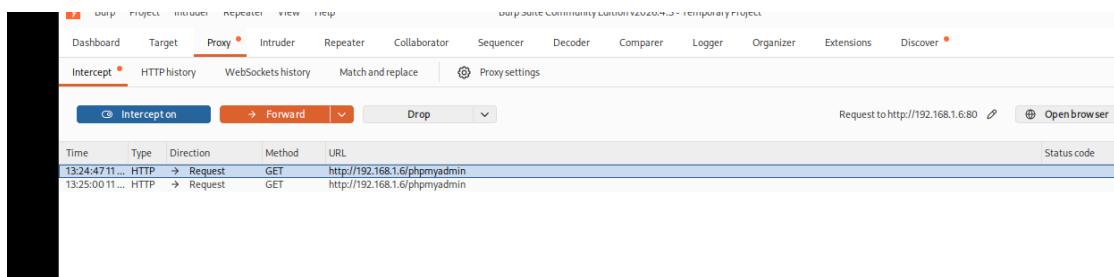


Figure 3: Burp Proxy 'Intercept' tab capturing live GET requests to the target host 192.168.1.6.

3.4 Manual Crawl of the Target Application

With Intercept switched off, the bWAPP login page was browsed normally using credentials bee / bug. Multiple pages were navigated — portal, training, password change, reset, and user management — allowing Burp to passively log the full request history into the Site Map.

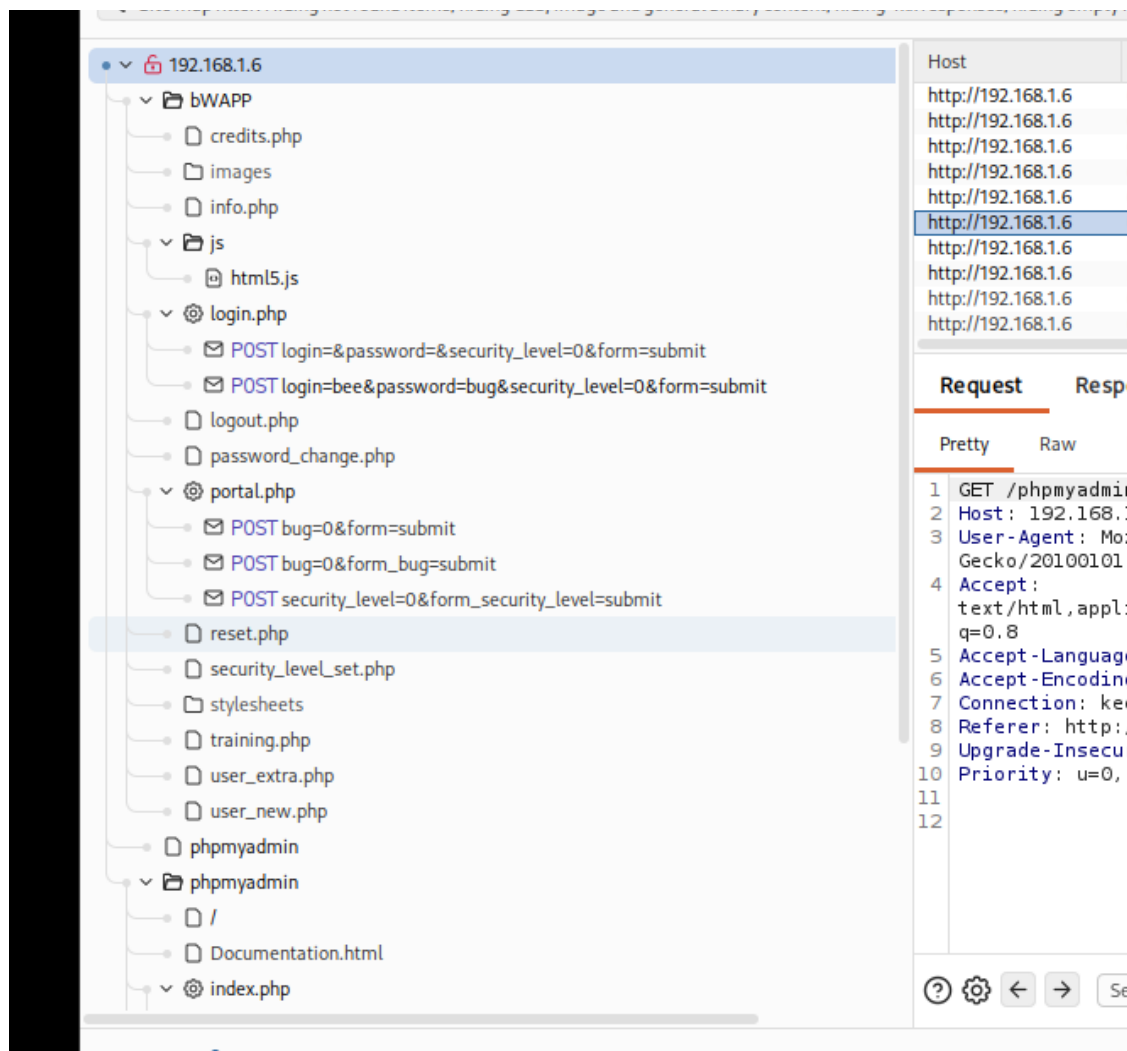
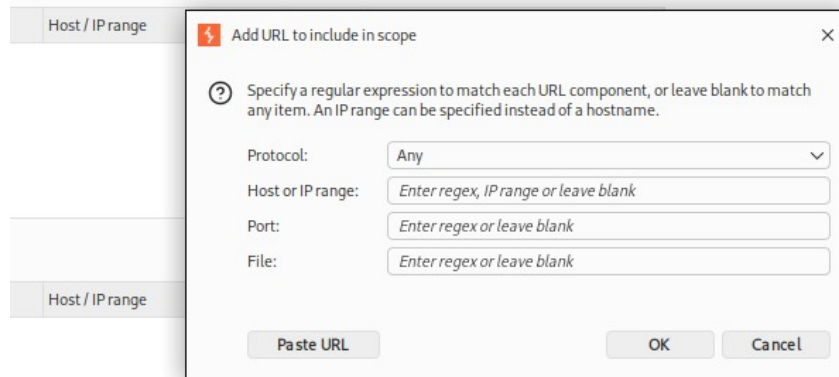


Figure 4: Site Map tree after manual crawling, showing directory structure and captured login requests.

3.5 Defining Target Scope

To focus analysis on the target only, the host 192.168.1.6 was added to Burp's Target Scope, and logging of out-of-scope proxy traffic was disabled to keep the Site Map clean.

ind URLs constitute the target for your current work. This configuration affects the behavior of tools throughout the suite.



Memory: 147.9MB of 1.90

Figure 5: 'Add URL to include in scope' dialog with the target host added.

3.6 Applying the Site Map Filter

The Site Map filter was opened and 'Show only in-scope items' was enabled under Filter by request type, ensuring the Site Map displayed only target-related content.

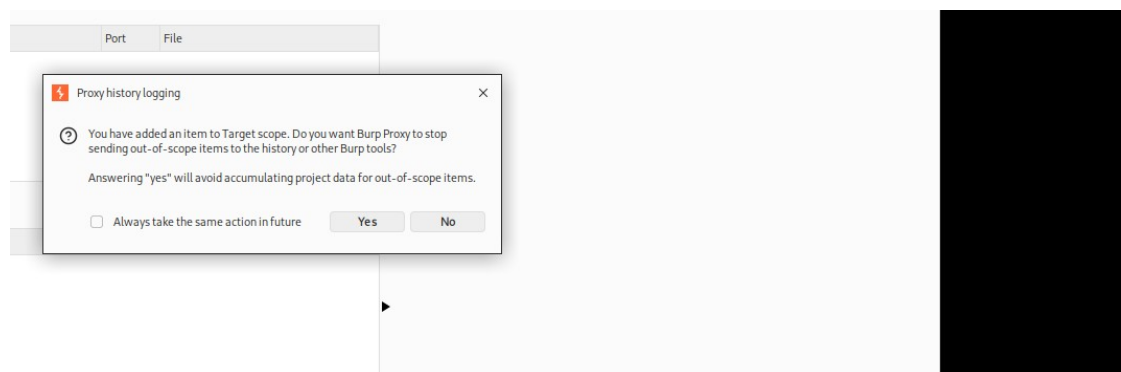


Figure 6: Site Map Filter dialog with options for request type, MIME type, status code, and folders.

3.7 Verifying Scope Filtering

After applying the filter, Burp displayed a confirmation banner indicating that logging of out-of-scope Proxy traffic had been disabled, verifying only target-related traffic would appear going forward.

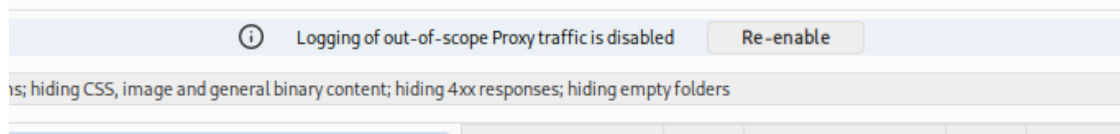


Figure 7: Confirmation banner — 'Logging of out-of-scope Proxy traffic is disabled'.

3.8 Reviewing Captured Requests

The expanded Site Map was reviewed, showing captured GET and POST requests for pages such as portal.php, reset.php, training.php, and user_new.php, along with parameters, status codes, and response lengths.

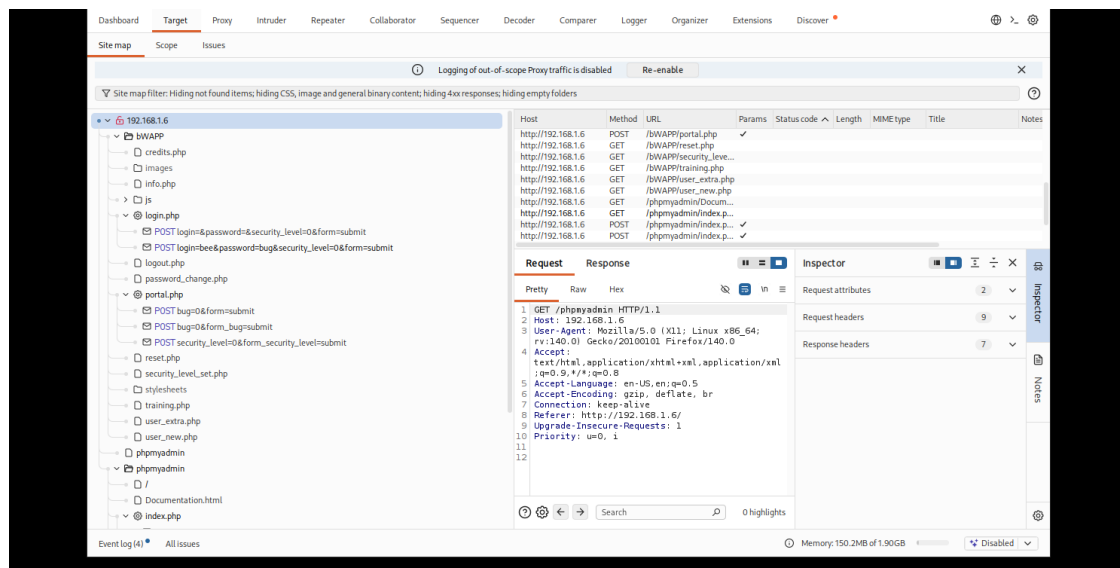


Figure 8: Scoped Site Map showing the directory tree alongside captured requests, parameters, and status codes.

4. Final Crawl Results

The screenshots below represent the completed state of the crawl — a fully scoped Site Map of the target bWAPP/bee-box application, showing the discovered directory structure and captured requests, cookies, and headers for individual pages.

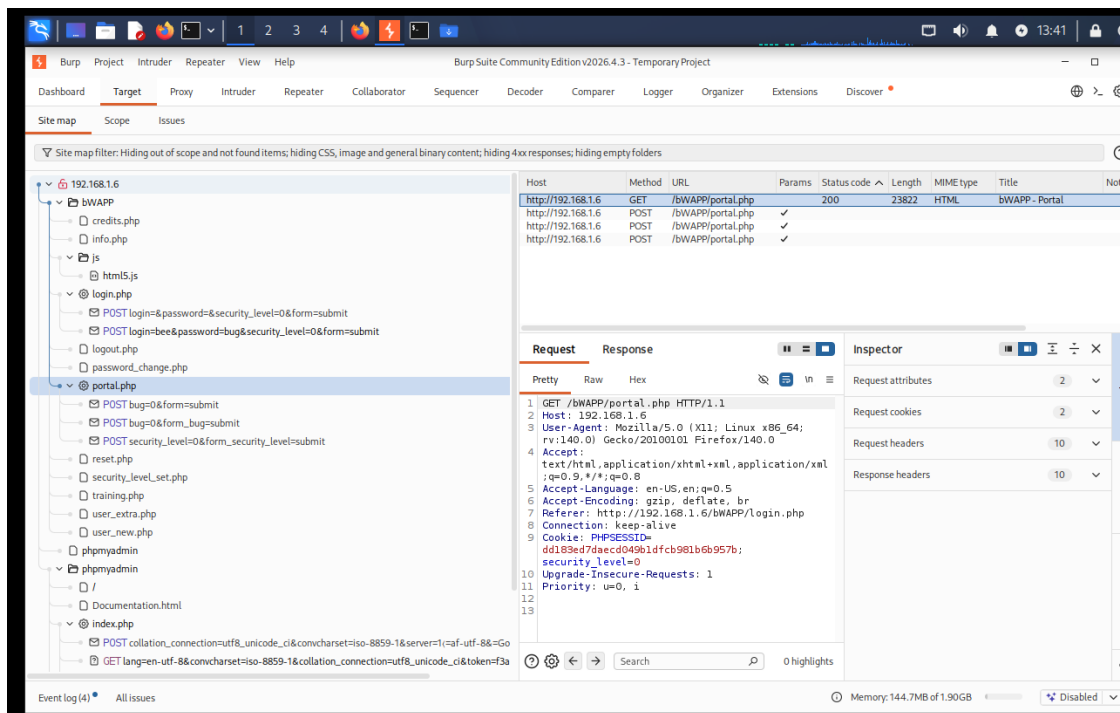


Figure 9: Final Site Map — target directory tree with request details for portal.php (session, cookies, headers).

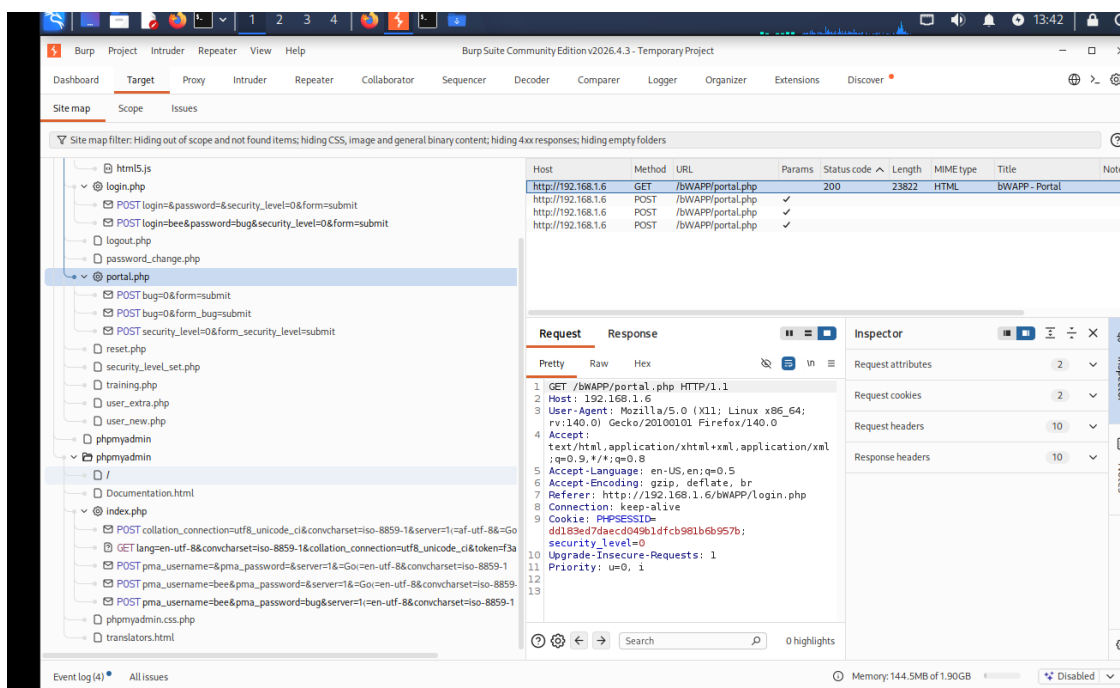


Figure 10: Final Site Map — lower section of the directory tree, including phpMyAdmin login requests and translation resources.

5. Directory Structure Identified

- /bwAPP/ — main application directory (login.php, portal.php, logout.php, password_change.php, reset.php, training.php, user_new.php, user_extra.php, security_level_set.php)
- /bwAPP/js/ — JavaScript resources (html5.js)
- /bwAPP/stylesheets/ — CSS resources

- /phpmyadmin/ — database administration interface (index.php, Documentation.html, phpmyadmin.css.php, translators.html)

6. Note on Automated Scanning

The automated crawl/scan workflow (New Scan dialog, automatic crawling, issue reporting) is a Burp Suite Professional-only feature. As Professional access required a business email address that was unavailable, this exercise was completed using Community Edition's manual crawling workflow instead — browsing the target through the intercepting proxy and building the Site Map organically — which achieves an equivalent result for a target of this size.

7. Conclusion

Web crawling of the bWAPP application hosted on the bee-box VM was successfully performed using Burp Suite Community Edition. The target's directory structure, application pages, and request/response data were fully mapped and scoped in the Site Map, providing a foundation for subsequent vulnerability analysis (SQL Injection, XSS, and authentication testing) in later stages of practice.

What I practiced in this exercise

- Configuring Burp Suite as an intercepting proxy and routing browser traffic through it.
- Defining and applying target scope to keep analysis focused and noise-free.
- Building a Site Map organically through manual crawling when automated crawling wasn't available.
- Reading captured requests/responses to understand an application's structure before testing it.